

GSA Gateway

A precision-first retrieval architecture for institutional question answering

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ABSTRACT

GSA Gateway is a self-hosted conversational assistant that answers questions about NJIT — its people, departments, research, programs, policies, and Graduate Student Association services — through Discord, Telegram, and GroupMe. It is built on one asymmetry: in an institutional setting a confident wrong answer is worse than no answer, because a student who is told the wrong deadline acts on it. The architecture follows from that premise. Questions that can be answered exactly are answered by deterministic SQL over a knowledge graph rather than from model recall, so the facts are read from the database, never invented. For sixteen of those skills the answer text bypasses the language model entirely — citation counts, URLs, and disambiguation rosters are rendered verbatim — and the rest are only rephrased by a local model under a check that no retrieved fact may be dropped. Generated prose is separately verified against the evidence that produced it, and the system abstains when support is absent. All model inference runs on local hardware and no question text reaches a commercial model provider; on a knowledge-base miss the question is sent to a commercial web-search API, scoped to njit.edu, to locate a source page.

27,685 INDEXED PASSAGES (~11,060 SERVED)	1,130 PEOPLE IN THE GRAPH	3,200 GRAPH EDGES	32 ROUTED SKILLS	92% ANSWERED (324-Q RUN)	2% JUDGED WRONG (OF ANSWERED)
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1 Design premise

A general-purpose RAG assistant optimizes for answering. That objective is wrong for a university help desk, where the cost of errors is asymmetric: an unanswered question sends a student to a web page, while a fabricated office number, deadline, or advisor name sends them somewhere that does not exist. GSA Gateway therefore optimizes for *precision under an abstention option*, and three rules follow.

First, prefer a deterministic path. A large share of real traffic — who chairs a department, which faculty work on a topic, a professor's citation count, how to reach an office — is a database query wearing a natural-language costume. Those questions are resolved by parameterized SQL against a knowledge graph, then either rendered verbatim or rephrased by a local model under a check that no retrieved fact is dropped — so the model never originates the fact. **Second, never let the model supply a fact retrieval did not.** Where a skill can tell that an asked-for attribute is missing, it states the gap rather than letting the model fill it; this is implemented skill by skill rather than as a blanket property. **Third, abstain over guess.** Generated answers are verified against their own source passages before delivery; unsupported ones are replaced by a live lookup or an honest non-answer.

2 Knowledge representation: two layers, one database

The system stores knowledge in two complementary layers inside a single SQLite database (WAL mode, STRICT tables), because institutional questions come in two shapes and neither representation serves both well.

The **text layer** holds 27,685 active passages — profile prose, policy pages, catalog entries, PDFs, curated GSA content, and 15,745 one-title publication records that are indexed but held out of the default answer corpus, leaving roughly 11,060 passages reachable by a general question. Each is indexed twice: an FTS5 full-text index over a generated `search_text` column, and dense vectors in a `sqlite-vec` virtual table. Embeddings come from Qwen3-Embedding-0.6B served locally by Ollama, 1024-dimensional and L2-normalized, using an asymmetric prefix scheme in which passages are embedded raw and only the query carries an instruction wrapper. Vector coverage of active passages is complete.

The **graph layer** holds 1,130 active people, 61 organizations, and 1,666 research areas as typed nodes, joined by 3,200 active edges: `has_role` (carrying a role category and the literal titles), `researches`, and `part_of`. Organization nodes bridge to a separate organizational tree; skills that rank or aggregate walk that whole subtree, so "most cited in the engineering college" spans its departments, while roster skills deliberately scope to the exact organization named. This layer is what makes exact answers possible: "who are the Computer Science faculty" is a traversal with a correct, complete, checkable answer, and treating it as a passage-similarity problem would be a category error.

3 The answer path

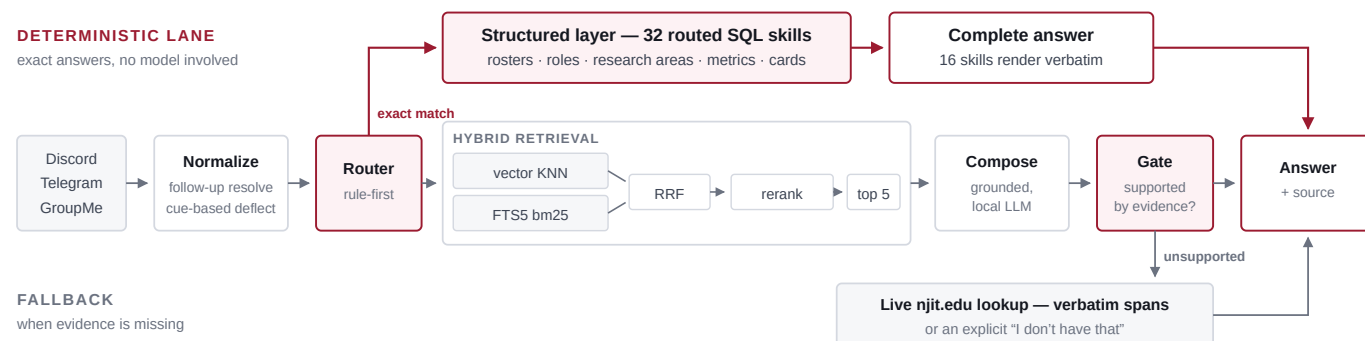


Figure 1 — Request flow. Anything the router can resolve exactly takes the deterministic lane, where 32 SQL skills produce a complete answer and 16 render it verbatim without a language model. Everything else takes the retrieval spine, where composed prose must be certified by a gate against its own sources; a failed gate does not guess.

4 Retrieval mechanics

Both search legs run over the same corpus — the default answer corpus, which holds out per-paper publication and syllabus rows — and are combined rather than chosen between. The semantic leg is a KNN scan of the `sqlite-vec` index; because all vectors are L2-normalized, Euclidean distance is converted to a cosine-equivalent similarity in closed form. The lexical leg is FTS5 with bm25 ranking, which is what actually retrieves exact identifiers — a course code, a surname, an acronym — that embeddings blur. The two ranked lists are merged by **Reciprocal Rank Fusion** ($k = 60$), which combines by rank position and so needs no score calibration between two incomparable scales.

The fused pool of 60 candidates is then reranked by a local ONNX cross-encoder (`ms-marco-MiniLM-L-6-v2`) that scores each query–passage pair jointly rather than comparing two independently-computed vectors. Its ranking is fused back with the lexical-semantic ranking using a deliberately asymmetric constant — 60 for the fused leg against 10 for the cross-encoder leg — so the cross-encoder dominates ordering while an exact keyword match retains a floor it cannot be reranked out of. Reranking is additive-safe: any failure to load or score returns to the pure fusion order rather than degrading the result.

A multiplicative prior then encodes freshness and document class before the final passages are selected: five entity-diversified primaries, plus the profile card of each person appearing among them, injected as identity context. News decays by age on a 180-day half-life but is floored, so an old announcement is demoted and never erased; undated news does not decay at all rather than being penalized for missing metadata; upcoming events are boosted while past ones are not; generic web pages are served at a reduced prior. Results are finally diversified by entity, so one heavily-documented professor cannot occupy every slot.

5 Correctness machinery

Retrieval quality alone does not prevent fabrication, so three independent mechanisms constrain what can reach the user.

Deterministic rendering. Of the 32 skills the router can reach, 16 skip language-model composition altogether. These are the outputs where rewording is itself a defect: citation counts and h-indices, external profile URLs, "which of these three people did you mean" rosters, and role titles whose qualifiers — *interim*, *joint appointment*, *affiliated* — carry meaning that a paraphrase would erase. Profile links and Scholar metrics are appended to the finished answer *after* composition and are never placed in the model's context, which makes a hallucinated URL or invented citation count structurally impossible rather than merely unlikely.

Honest-partial routing. When a question asks for an attribute the data does not cover for the entities involved, the system reports the gap instead of filling it. Asked for the research areas of a department's faculty, it names only those who actually list areas, says how many faculty that is, and degrades to a plain roster prefaced by an explicit "I don't have research areas listed" statement when none do. The earlier behavior — inventing one plausible topic per name — is precisely the failure this rule exists to prevent.

The abstention gate. Generated RAG answers pass through a two-stage check. A high-precision deterministic matcher first deflects questions the system cannot legitimately answer at all — requests about a user's own account, records, or another institution — before retrieval spends effort on them. After generation, a deterministic stage re-reads the answer against the passages that produced it: a self-decline check, a subjective-superlative guard, and a requirement that any count, rate, amount, or date stated in the answer also appear in the evidence. Only the residual costs a zero-temperature model call, which judges whether the retrieved passages answer the question at all and must return a supporting quote, itself then grounding-checked against those passages. A rejected answer is not shipped: the system tries the live tier, and failing that returns a useful non-answer. The gate is deliberately fail-open — a fault in the checker keeps the composed answer rather than silently suppressing it — and it does not run on small-talk turns. On a frozen 45-question evaluation, replacing the previous production gate with this one moved the false-answer rate from 40% to 15% while holding

false abstentions roughly flat (24% to 20%); the same evaluation scored an ungated pipeline at 65%. The gate covers the generative path; among the structured skills, 16 bypass the model entirely and the other 16 are rephrased from complete deterministic facts under a fact-preservation check that reverts to the verbatim facts whenever a counted roster or an appointment qualifier would be dropped.

6 Data acquisition and operation

Knowledge enters through six producers, each tagged with its own provenance so that a routine re-crawl does not disturb another producer's rows. A **people crawler** walks 17 anchored entry points across NJIT's colleges and builds the graph — people, roles, organizations, research areas — from server-rendered profile pages. A **sitemap-driven site crawler** covers 47 university subsites and supplies the bulk of the prose corpus. A **department prose crawler** walks college subdomains directly, a **catalog crawler** ingests course and program records, and **Scholar enrichment** attaches citation metrics to 363 of the 1,130 active people, each matched to a verified Scholar profile. Graduate Student Association and student-club content is authored by hand through a local dashboard, because its source is a site that cannot be crawled.

Ingestion is deliberately mechanical. Crawlers may strip markup, navigation, and encoding noise, but they may not summarize, paraphrase, or reword: the human-readable text passes through unchanged, the one exception being a 6,000-character cap on captures of faculty personal sites, and every decision about how content is *used* belongs to the retrieval layer. This separation is what allows NJIT source text to be served verbatim with a link, rather than as a model's recollection of it.

The prose crawlers, the Scholar refresh, and every one-off migration are gated: they default to a dry run, and promoting one to a commit takes a hardened snapshot first — the SQLite online-backup API, integrity-checked and rotated. The people crawler and the embedding pass are the exceptions, writing the live database directly with the snapshot available as an explicit flag. Per-entity reconciliation after a re-crawl is scoped by producer, so refreshing crawled data does not retire hand-authored records or enrichment sharing an entity; the deliberate exception is departure, where a person who has lost every appointment has all of their passages retired regardless of producer. For the sitemap-driven crawlers, retirement is further guarded by a floor — a frontier below 300 URLs, or below 80% of the prior active set, skips the retirement pass entirely — so a partial fetch cannot mass-delete a corpus. Embedding is a separate resumable pass that only processes rows lacking a vector.

7 Evaluation

The system is measured by replaying a maintained question suite — currently 336 questions across 79 categories, extended whenever a change ships — through the real pipeline rather than a test harness, then classifying each response and judging it with a model grader. The most recent full judged run, from 9 July 2026 against a 324-question snapshot of the suite, scored:

OUTCOME	COUNT	SHARE	READING
Answered from the knowledge base	244	75%	Served locally, no external call
Answered from a live njit.edu lookup	55	17%	Coverage gap, correctly recovered
Declined	25	8%	Deliberate abstention
Fully correct (of the 299 answered)	215	72%	Complete and accurate
Partially correct	78	26%	Grounded but incomplete
Wrong	6	2%	The number the design targets

The shape of that distribution is the point. Outright wrong answers sit at 2%, while the dominant residual failure is partial coverage — answers that are true but incomplete. That is the trade the architecture is explicitly making, and incompleteness is a retrieval problem that more data fixes, whereas fabrication is a trust problem that no amount of data fixes. The suite has since grown to 336 questions and has not been re-judged, so these figures describe the system as of that run.

8 Limitations

- **Partial answers are the standing weakness.** Roughly a quarter of answered questions are judged incomplete rather than wrong — usually a question with several clauses where retrieval covers some.
- **News coverage is thin.** The corpus holds 88 news rows, mostly index pages rather than articles, and only one carries a publication date; the university newsroom is not crawled. News questions are effectively carried by the live-lookup tier at query time rather than answered from local knowledge.
- **Ingestion is operator-triggered, not scheduled.** Every crawl, refresh, and embedding pass is run by hand under the gated workflow. Freshness is therefore a function of operator attention; full automation is a design goal, not a shipped property.
- **The gate is calibrated to the current models.** Its thresholds were tuned against the deployed generation and embedding models on a 45-question set. Swapping either model requires re-embedding the corpus and re-validating the gate; the measured figures should not be read as a general guarantee.

- **Enrichment coverage is uneven.** 363 of 1,130 people carry verified Scholar metrics, because the source disallows automated collection and matching is held to a strict verification standard rather than a best guess.

9 Implementation

LAYER	CHOICE	RATIONALE
Generation	IBM Granite 4 (<code>granite4:tiny-h</code>) via Ollama	Runs locally on modest hardware; composition is constrained to supplied facts, so raw model capability is not the binding constraint
Embeddings	Qwen3-Embedding-0.6B, 1024-d	Local, asymmetric query/passage scheme; the model is a swappable descriptor, not a hard-coded assumption
Reranking	ONNX cross-encoder, CPU	Joint query–passage scoring; fails safe to fusion order
Storage	SQLite (WAL, STRICT) + <code>sqlite-vec</code> + FTS5	One file holds passages, vectors, keyword index, and graph — no separate vector service to operate or keep consistent
Surfaces	Discord, Telegram, GroupMe	All conversational; the only commands are <code>/qrcode</code> and Telegram's mandatory <code>/start</code>
Operations	Local dashboard on <code>127.0.0.1:5555</code>	Content editing and job control, reachable only over an SSH tunnel

Beyond the messaging platforms themselves, the only outbound dependencies in the answering path are the search API and the `njit.edu` page fetch used by the live-lookup tier when local knowledge misses. Generation, embedding, and reranking are entirely local, user identifiers are hashed before any write, and question text never reaches a commercial model provider. The codebase is approximately 97,000 lines of Python with 431 test files, developed over 1,182 commits since May 2026.

GSA Gateway · Graduate Student Association, New Jersey Institute of Technology · Maintainer: Mohammad Dindoost (`md724@njit.edu`). Figures reflect the deployed system as of 22 August 2026 and were verified against the running database and source at the time of writing. Evaluation figures come from the project's own replay harness; the frozen 45-question gate evaluation and the 324-question judged run are separate measurements and are labeled as such.